



FIREBRAND

LYCEUM EDUCATION GROUP

Iteration and Data Handling in Power Platform

Power Platform Pathway | Session 3

Session Recording & Professional Expectations

Session Recording

- All Estio training sessions are recorded to:
 - Support learner progress
 - Enable review of content
 - Maintain quality and compliance standards
- By attending this session, you acknowledge that recording is taking place.

Camera & Participation Expectations

- To support engagement and attendance monitoring:
 - Cameras should remain on during the session
 - Please participate actively in discussions and activities
 - If you are unable to use your camera or microphone for any reason, please inform your tutor directly via chat or message.

This session forms part of your professional apprenticeship experience. Please engage respectfully and professionally throughout.

Safeguarding, British Values & Wider Learning

Safeguarding protects individuals from harm, abuse, neglect, radicalisation and exploitation.

If you have a concern about yourself or someone else, email:

safeguarding@bpp.com

Or speak to your tutor privately.

We treat all concerns seriously and confidentially.

Safeguarding



The British Values are:

1. **Democracy**
2. **Rule of Law**
3. **Individual Liberty**
4. **Mutual Respect**
5. **Tolerance of Different Faiths and Beliefs**

These values underpin everything we do in professional and digital environments.

British Values



Prevent is about protecting people from being drawn into extremism or radicalisation.

In our programmes, Prevent links to:

- **Online safety**
- **Digital responsibility**
- **Critical thinking**
- **Professional ethics**

Prevent



As we go through today's session, consider:

How today's topic connects to British Values

Whether there are any Prevent or digital safety themes

How this learning supports your professional and career development

Your tutor will guide discussion and help you make these connections.

Competencies and Coverage

Professional Discussion underpinned by Portfolio

Knowledge

K1, K3, K4, K5, K7, K8,
K10, K12

Skills

S2, S3, S5, S8, S9, S13,
S14, S15, S17

Behaviours

B1, B4, B5, B6, B7, B8,
B9

Work-based Project with Questioning

Knowledge

K2, K6, **K9**, K11

Skills

S1, S4, S6, **S7**, S10,
S11, S12, **S16**

Behaviours

B2, B3

Lesson Aim and Objectives

Lesson Aim: Iteration and Data Handling in Power Platform

Lesson Objectives

Apply

- Apply iteration techniques using collections

Analyse

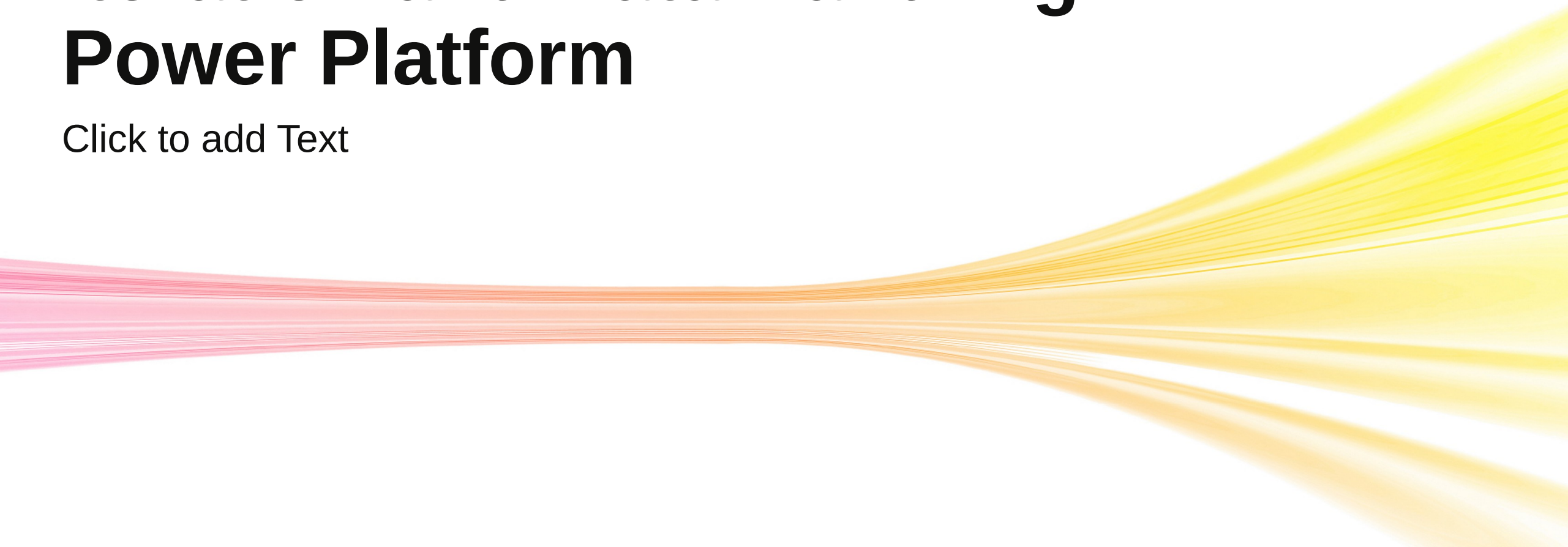
- Analyse how data is processed within apps

Evaluate

- Debug issues in data handling and logic

Iteration and Data Handling in Power Platform

Click to add Text

A decorative graphic consisting of several overlapping, wavy lines that originate from the left side of the slide and extend towards the right. The lines are colored in a gradient of pink, orange, and yellow, creating a sense of motion and energy.

Introduction to Iteration

What is Iteration?

- **Repeated execution:** Running code blocks multiple times
- **Core block:** Essential foundation for algorithm design
- **Efficiency:** Eliminates redundant, copy-pasted code statements

The Two Loop Types

- **Definite iteration:** Loops run a predetermined number of times
- **Indefinite iteration:** Loops run until a condition changes

Collections

Core Mechanics:

- A collection is an in-memory table of records, similar to a list of dictionaries
- Collect() adds records; ClearCollect() clears and rebuilds a collection in one step

ClearCollect(

colAttendees,

{ Name: "Priya", Present: true }

);

Collect(colAttendees, { Name: "Sam", Present: false })

ForAll() - Processing Every Record

Core Mechanics:

- ForAll() runs a formula once for every record in a table or collection

```
ClearCollect(  
    colSummary,  
    ForAll(  
        colAttendees,  
        { Name: Name, Status: If(Present, "In", "Out") }  
    )  
)
```

Aggregation Functions

Core Mechanics:

- CountRows(), Sum(), Average(), Max() and Min() summarise a table without a manual loop

// Label.Text property

"Total attendees: " & CountRows(colAttendees)

Efficient vs. Inefficient Data Handling

Inefficient - Rebuild on Every Keystroke:

// OnChange of every input

`ClearCollect(colAttendees, ...)` // wipes and rebuilds constantly

Efficient - Update Only What Changed:

// Button.OnSelect

`Collect(colAttendees, { Name: NameInput.Text, Present: true })`

Why does rebuilding the whole collection on every keystroke hurt performance?

- **Overworks Memory:** Wiping and recreating a data table for every single letter typed strains the device's RAM
- **Causes Interface Lag:** Forces connected layout galleries to constantly redraw, causing screen flickering and stutter
- **Drops Mobile Performance:** Drains battery and slows down processing speeds heavily on mobile screens



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Guided Hands-On Exercises

Guided Data Handling Activity



ACTIVITY



Structured Practice with Collections:

- Create a collection from user input controls
- Add records dynamically using buttons and formulas
- Bind the collection to a gallery for display

Tasks:

- Update the collection and observe changes in the UI
- Describe the data flow from input to collection to UI output

Expected Outcome: Working Power Apps solutions demonstrating correct creation, updating and display of collection-based data



ACTIVITY



Next Activity will begin soon



Attendance Tracker App



ACTIVITY



Objective: Build an app that captures and summarises attendance

Tasks / Requirements:

- Capture attendee names into a collection
- Display records within a gallery
- Use CountRows() to display a dynamic total

Key Point: Gallery controls must update dynamically as the collection changes

Extension Activity: Add validation to prevent duplicate or blank entries



ACTIVITY



Next Activity will begin soon



Data Processing Challenge



ACTIVITY



Working with a Provided Dataset:

- Process records using ForAll() or filtering functions
- Calculate totals and averages

Tasks:

- Display results clearly within the user interface
- Identify simple patterns and trends within the data

Expected Outcome: Accurate data processing demonstrating effective use of Power Apps formulas and collection-based operations



ACTIVITY



Data Processing Challenge (Dataset)



// 1. Paste this into Screen.OnVisible to load the challenge data:

```
ClearCollect(colSalesRecords,  
    { ID: 1, Product: "Mice", Category: "Hardware", TotalGross: 150 },  
    { ID: 2, Product: "Office 365", Category: "Software", TotalGross: 1200 },  
    { ID: 3, Product: "Keyboards", Category: "Hardware", TotalGross: 350 }  
);
```

// 2. UI Formula Hints:

// Calculate Gross Sales: Sum(colSalesRecords, TotalGross)

// Filter for Hardware Only: Filter(colSalesRecords, Category = "Hardware")

Tracing Data Flows

The 3-Point Audit: When a user interface element behaves unexpectedly, trace the issue sequentially through these three distinct layers:

1. **INPUT LAYER** → Is the raw value being passed in the correct format (e.g., Text vs. Numeric)?
 2. **STORAGE LAYER** → Open the **Variables / Collections panel** via the side menu. Is the table populating in real-time RAM?
 3. **OUTPUT LAYER** → Inspect the target control property. Is it pointing to the correct column name mapping?
- **The LookUp Validation Rule:** To catch duplicate record bugs, always verify that your query structure searches *existing rows* before executing append commands
 - **Schema Sensitivity:** Power Fx columns are strictly typed and case-sensitive. Mapping *ThisItem.name* instead of *ThisItem.Name* will instantly blank out layout fields

Debugging and Refinement Activity



ACTIVITY



Common Issues Provided:

- Collections not updating correctly
- Duplicate records being stored
- Incorrect gallery field bindings
- Formula errors referencing incorrect fields

Tasks:

- Trace the complete flow of data through the application
- Identify and annotate the root cause of each issue
- Correct the problem and retest the app & verify that the UI displays accurate data

Expected Outcome: Evidence of a structured debugging process that considers data flow, formula logic and UI behaviour



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Evaluation Discussion



Strategic Collection Architecture

The Efficiency Rule

When to Collect

When to Avoid

Performance Impact

Maintainability Impact

Collection Efficiency & Impact

The Golden Rule:

Collections are **in-memory arrays**

Every row added consumes local device hardware RAM

Best Architecture Scenarios:

- *Staging Items*: Local shopping carts, purchase order line-items before checkout
- *Offline Forms*: Storing records when a device has no Wi-Fi, then patching them later
- *Caching*: Caching stationary data (like a country dropdown list) once at startup to avoid repeated database calls

• The Anti-Pattern Network Fall:

Do not use collections as a clone of a large database table

They break delegation and cause long app loading screens

• Maintainability Risk:

Spreading complex `Collect()` formulas across multiple screens makes data flows impossible to trace during an outage

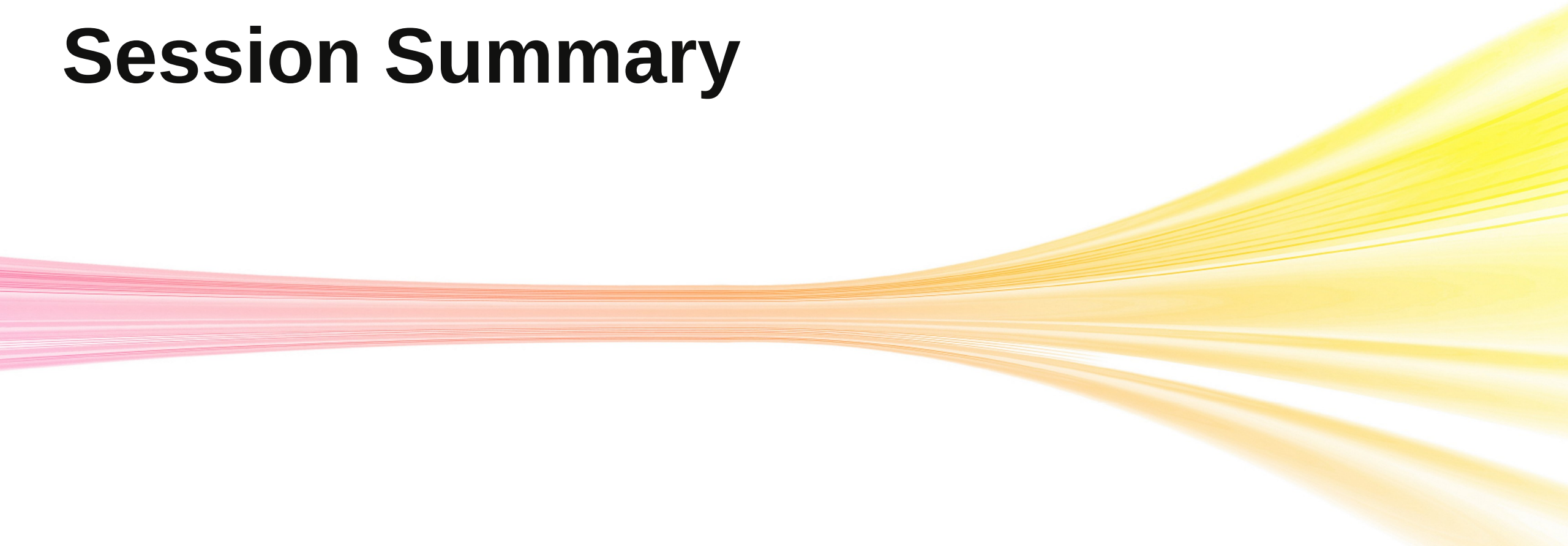
Stretch and Challenge

The Mission:

Improve your solutions by doing the following:

- Restructure data flow to reduce unnecessary processing
- Reduce redundant collection operations
- Improve formula efficiency

Session Summary



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